



RS004 Week 11 — English Worksheet

Topic: Coins & Score · **Course:** Platformer Game · **Time:** about 45 minutes

This week is about **reading and explaining** the coin code you wrote in your live lesson. You keep coins in a list, check if the player touches one, **remove** it, and add to the **score**. Then you draw the score on screen with `pygame.font`. Here you think about how that code works and explain it in your own words.

Keep these words handy: **coin**, **collect**, `remove()`, **score**, **HUD (on-screen text)**.

1 · Predict 🧠

Read each snippet. Write what you think it does.

```
for coin in coins[:]:
    if player.colliderect(coin):
        coins.remove(coin)
        score = score + 1
```

The player touches a coin. What two things happen?


```
score_text = font.render(f"점수: {score}", True, (0, 0, 0))
screen.blit(score_text, (10, 10))
```

Where on the screen does the score appear? What does `render` make?


```
for coin in coins[:]:
```

Why loop over `coins[:]` (a copy) instead of `coins` while removing items?

2 · Trace the Score

The player collects coins one by one. Fill in the score after each step. Start with `score = 0` and 3 coins.

```
collect coin 1:  score = ____  coins left = ____  
collect coin 2:  score = ____  coins left = ____  
collect coin 3:  score = ____  coins left = ____
```


3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — Touching a coin should remove it and add a point. Right now the score goes up but the coin stays, so it scores forever.

```
for coin in coins[:]:
    if player.colliderect(coin):
        score = score + 1
```

Hint: the coin is never removed from the list.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Removing coins while looping over the original list skips some coins and may crash.

```
for coin in coins:
    if player.colliderect(coin):
        coins.remove(coin)
        score += 1
```

Hint: loop over a *copy* of the list while you remove from the real one.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The score should show on screen, but nothing appears even though `score` is going up.

```
score_text = font.render(f"점수: {score}", True, (0, 0, 0))
```

Hint: rendering the text isn't enough — you must `blit` it onto the screen.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code

Read this working coins-and-score code, then answer the questions about it.

```
coins = [pygame.Rect(200, 300, 16, 16), pygame.Rect(420, 180, 16, 16)]
score = 0
font = pygame.font.SysFont(None, 36)

for coin in coins[:]:
    if player.colliderect(coin):
        coins.remove(coin)
        score = score + 1

score_text = font.render(f"점수: {score}", True, (0, 0, 0))
screen.blit(score_text, (10, 10))
```

What kind of data structure holds the coins, and what is each coin?

What does `player.colliderect(coin)` check, and when is it `True` ?

What two things happen inside the `if` when a coin is touched?

Why does the loop use `coins[:]` instead of `coins` ?

What is the difference between `font.render(...)` and `screen.blit(...)` ?

5 · Explain Your Lesson Code 🎥

Explain the coins-and-score code **you wrote in today's lesson**. Record a short phone video. You may show it running. Try to use these words: **coin, collect, remove, score, render**.

1. Show your coins placed in the level.
2. Collect a few and show the score climbing.
3. Read your collect-and-remove lines out loud and explain them.
4. Explain how your score gets drawn on screen.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.